



HWA CHONG INSTITUTION (COLLEGE SECTION)
2018 JC2 9744 H2 BIOLOGY
PRELIMINARY EXAMINATION PAPER 3 MARK SCHEME

SECTION A

QUESTION 1

(a) (i) Describe how the structure of glycogen is related to its function. [2]

- Ref. to glycogen being a large molecule and hence allowing storage
- Ref. to specific bonds in glycogen and their relevant function of either storage or easy hydrolysis

(ii) explain the significance of localising GAA into the lysosomes. [3]

- Ref. to idea of lysosome being an acidic environment, hence optimal GAA
- Ref. to idea of compartmentalisation
- Ref. to idea of regulated hydrolysis of glycogen

(b) (i) Account for the results shown in Fig. 1.2. [4]

- Ref. to idea of glycogen deposit in healthy and diseased individuals
- Ref. to appropriate supporting quote
- Ref. to idea of diseased individuals having significant glycogen deposits in lysosomes
- Ref. to appropriate supporting quote

(ii) Suggest how the mutated GAA gene results in cellular damage. [2]

- Ref. to idea of low concentration of GAA enzymes
- Ref. to relevant consequence resulting in cellular damage

(c) (i) With reference to Fig. 1.3, explain why Pompe disease is an autosomal recessive disease. [3]

- Ref. to why Pompe is autosomal
- Ref. to appropriate supporting quote
- Ref. to why Pompe is recessive
- Ref. to appropriate supporting quote

(ii) None of the children of individuals **III-1** and **III-2**, other than individual **IV-4**, suffered from early onset Pompe. Suggest in whom and in what organ a mutation occurred that accounts for this. [2]

- Ref. to either individuals III-1 / III-2
- Ref. to gonads

(d) Explain how genotype is linked to phenotypes in Pompe disease. [3]

- Ref to idea of different mutations present
- Ref to idea of different levels of GAA enzyme activity
- Reference to appropriate quotation from table 1.1 to support conclusion

(e) (i) Explain how ERT works to treat Pompe individuals. [4]

- Ref. to idea of Pompe individuals absent GAA enzymes
- Ref. to transport of active GAA in blood to cells
- Ref. to endocytosis and fusion with lysosomes
- Ref. to reduction in symptoms in diseased individuals

(e)(ii) Suggest why Pompe individuals require weekly infusions of active GAA enzymes. [2]

1. Ref. to idea of degradation of active GAA enzymes
2. Ref. to idea of the need to maintain minimum levels of active GAA enzymes to prevent symptoms of Pompe

[Total: 25]

QUESTION 2

- (a) Explain why there is a need to mix isolated bacteriophages with pathogenic bacteria. [2]
- Ref to bacteriophages causing lysis of the pathogenic bacteria
 - Ref to bacteriophage multiplication
- (b) With reference to its reproductive cycle, suggest how a virulent bacteriophage could be used to treat bacterial infections. [5]
- Tail fibres attach to specific receptor sites
 - DNA is extruded
 - Halt synthesis of host DNA, RNA and proteins
 - Hijack host cell ribosomes and DNA polymerase to synthesise viral proteins and replicate viral DNA
 - Assembly of new virion particles
 - Lyse pathogenic bacteria thus treating the infection
- (c) Explain why bacteriophage therapy may be a better mode of treatment than antibiotics. [3]
- Bacteriophage is more specific
 - Overcomes problems of drug resistance
 - Does not kill natural microflora
- (d) Suggest and explain for the controversial nature of this form of bacteriophage therapy. [3]
- Ref to safety / ethical / social, concerns
 - Ref to the need to modify host bacteria which could make them more pathogenic
 - Ref to the possibility of mutations to target recipient body cells instead of bacteria cells

[Total: 13]

QUESTION 3

(a) Outline how penicillin acts on bacteria and use Fig. 3.1 to suggest why penicillin has little or no effect at treating diseases caused by Gram-negative bacteria, such as some strains of *E. coli*. [4]

- Penicillin weakens the cell wall by binding to and blocking transpeptidases
- Thereby preventing the formation of, cross links / linkages, between NAM residues in the transpeptidation step during cell wall synthesis
- The weakened cell wall cannot withstand the pressure potential and the cells eventually would burst.
- Presence of the outer membrane, prevents / interferes with / protects from, entry of penicillin.

(b) Using Fig. 3.1, suggest the possible roles of the periplasm in bacteria. [2]

- Periplasm is where peptidoglycan synthesis occurs
- Periplasm contain proteins for secretion / folding / oxidation
- Periplasm contain proteases / nucleases / phospholipases/ phosphatases

(c)(i) Define the term *lipopolysaccharide*. [2]

- Each lipopolysaccharide consists of a lipid and a polysaccharide chain made up of repeating units of monosaccharides
- Ref. to covalent bond between lipid and carbohydrate

(ii) Suggest and explain why infection with one pathogenic strain of *E. coli* does not provide immunity to a different pathogenic strain. [4]

- Ref to different strains of bacteria containing different antigens
- Ref to antigens composed of different sugars
- Upon infection with one pathogenic strain, the antigens will trigger an immune response whereby receptors of the lymphocytes are specific/complementary to the antigen of *E. coli*
- Ref to different antibodies synthesised for different antigens / O polysaccharides / lipopolysaccharides

[Total: 12]

SECTION B**QUESTION 4**

- (a) Eukaryotic chromatin, the DNA-nucleosome polymer, is a dynamic molecule that exists in different configurations during the course of the cell cycle.

Describe how DNA is packaged in a cell and discuss the advantages of the eukaryotic chromatin being a dynamic molecule. [14]

- DNA coiled around histone proteins
- Histones having positively-charged residues form ionic bonds with the negatively-charged sugar-phosphate backbone of DNA.
- with linker DNA, gives a “beads on a string” look/ 10nm chromatin fibre
- DNA is further folded or coiled to produce the 30-nm chromatin fibre/ solenoid.
- Histone H1 and linker DNA are involved
- scaffold proteins are involved
- in condensing the 30-nm chromatin fibre to form looped domains
- which further coil to produce condensed chromosome
- Allow the DNA to be more compact to fit into the nucleus
- Ref to ability to regulate gene accessibility
- Ref to the idea of allowing/ preventing, RNA polymerase, to gain access to promoter to turn genes off/on
- DNA does not get entangled and break mitosis/ meiosis
- Ref to enabling cell division / mitosis / meiosis to occur successfully
- Ref to uncondensed chromatin allows for ease of DNA replication

- (b) The study of identical twins provides an opportunity for exploring the extent to which epigenetics and environmental factors contribute to phenotypic variation in humans. Epigenetics is the study of heritable changes in gene expression that do not involve changes to the underlying DNA sequence. Epigenetic silencing turns gene off, contributing to differential gene expression in specialised cells.

Describe how epigenetic silencing may occur and discuss the extent to which epigenetic silencing accounts for identical twins being phenotypically non-identical. [11]

- DNA methylation, addition of methyl groups to cytosine nucleotides
- DNA methylation changes the 3D conformation of DNA
- Methylated DNA to recruit histone deacetylases
- Histone deacetylation, removal of acetyl groups of lysine residues in histone tails
- Lysine residues regain their positive charges, increase in the affinity of the histone complex for DNA
- Consequently, the chromatin becomes more compact
- Different genes may be switched on / off in one twin but are not expressed in the other
- Different environmental factors
- Diet and e.g of type of diet
- Difference in exposure to sunlight, resulting in difference in skin colour
- Difference in lifestyle/ exposure to mutagens

[Total: 25]

QUESTION 5

- (a) The flow of electrons in oxidation-reduction reactions is responsible, directly or indirectly, for all work done by living organisms. Plants generate cellular energy in the form of ATP by the same process of chemiosmosis.

Describe the process of chemiosmosis and discuss the importance of electron flow in both chloroplasts and mitochondria that leads to chemiosmosis, with particular emphasis on the role of membrane proteins. [14]

- Ref to a proton gradient is generated across a membrane
- Ref to synthesis of ATP
- Ref to ATP synthase
- Ref to energy is released
- Ref to proton pumps to actively pump H^+ into thylakoid lumen / intermembrane space
- Membranes are impermeable to H^+ leading to generation of proton gradient
- Ref to role of PSII / PSI in light harvesting
- Light energy excite an electron in PSII / PSI to higher energy
- Capture of electron by primary electron acceptor
- Electrons are passed down electron carriers through a series of redox reactions
- Flow of electron to the final electron acceptor NADP, reducing it to NADPH
- Ref to ETC embedded in the inner mitochondrial membrane contributes to the chemiosmotic gradient for ATP synthesis
- Ref. to NADH and $FADH_2$ being reoxidised, when they donate electrons / reducing the electron carriers
- Flow of electrons to final electron acceptor, oxygen to form water

QWC:

Clear and unambiguous communication of a breadth of ideas on the process of chemiosmosis as well as photosynthesis and cellular respiration.

- (b) In plants, carbon dioxide (CO_2) and water (H_2O) are the products of cellular respiration while they are the reactants in photosynthesis. Tropical rain forests are estimated to be responsible for approximately 20% of global photosynthesis, yet the consumption of large amounts of CO_2 by living trees is thought to make little or no net contribution to reduction of global warming.

Explain how photosynthesis and cellular respiration are related to the climate and discuss the contributing factors that continue to bring about global warming even with large amounts of CO_2 being consumed by trees in the tropical rainforests. [11]

- Greenhouse gases (GHGs) like CO_2 and CH_4 trap and absorb heat at the Earth's surface, keeping the Earth warm
- Ref to uptake of atmospheric CO_2 by photosynthetic organisms
- Ref to sugar molecules formation in the Calvin cycle
- Heterotrophs consume plants to obtain the energy trapped
- Heterotrophs break down sugars / other forms of respiratory substrates
- Ref to CO_2 being produced during link reaction and Krebs cycle
- Ref to the idea of anthropogenic factors, being man-made / actions of humans
- Deforestations leads to the cessation of CO_2 absorption
- Burning of fossil fuels due to increase in global demand for energy
- Ref to how agriculture / rice cultivation / landfill waste / food choices / increase in the consumption of meat contribute to global warming

- Leading to increase livestock farming / domestication of animals / ideal conditions for bacteria to decompose organic material / waste decomposes in absence of O_2 / enteric fermentation of ruminant animals
- Ref to drastic increase in amount of CO_2 and CH_4 emissions

QWC:

Scientific argumentation exemplified by linking coherently photosynthesis and cellular respiration to the climate as well as how causal factors bring about global warming.

[Total: 25]